

```

import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

from sklearn.cluster import KMeans

data = pd.read_csv('old_faithful_272.csv')

print(data.shape)
print(data.head())

(272, 2)
   eruptions  waiting
0      3.600      79
1      1.800      54
2      3.333      74
3      2.283      62
4      4.533      85

data.isnull().sum()

eruptions    0
waiting      0
dtype: int64

x = data[['eruptions', 'waiting']].values

print(x[:5])

[[ 3.6   79. ]
 [ 1.8   54. ]
 [ 3.333  74. ]
 [ 2.283  62. ]
 [ 4.533  85. ]]

wcss = []

for i in range(1, 11):
    model = KMeans(
        n_clusters=i,
        init='k-means++',
        random_state=42,
        n_init=10
    )

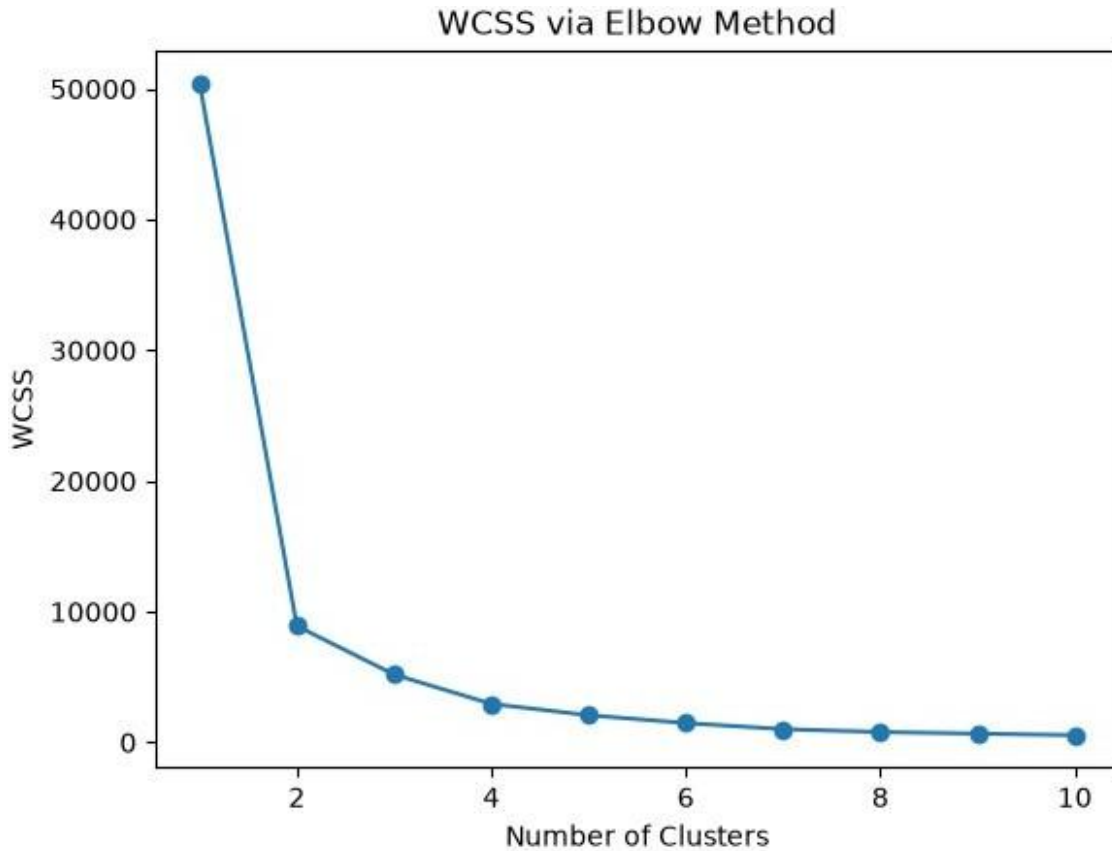
    model.fit(x)

    wcss.append(model.inertia_)

plt.plot(
    range(1, 11),

```

```
wcss,  
marker='o'  
)  
  
plt.title('WCSS via Elbow Method')  
plt.xlabel('Number of Clusters')  
plt.ylabel('WCSS')  
  
plt.show()
```



```
model = KMeans(  
    n_clusters=2,  
    init='k-means++',  
    random_state=42,  
    n_init=10  
)  
  
y_means = model.fit_predict(x)  
  
print("Cluster Labels:")  
print(y_means)
```

```

Cluster Labels:
[0 1 0 1 0 1 0 0 1 0 1 0 0 1 0 1 1 0 1 0 1 1 0 0 0 0 1 0 0 0 0 1 0 0
1 1
0 1 0 0 1 0 1 0 0 1 1 0 1 0 0 1 0 1 0 0 1 0 0 1 0 1 0 1 0 0 0 1 0 0 1
0 0
1 0 1 0 0 0 0 0 0 1 0 0 0 0 1 0 1 0 1 0 1 0 0 0 1 0 1 0 1 0 0 1 0 1 0
0 0
1 0 0 1 0 1 0 1 0 1 0 0 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 1 0 0 0 1
0 1
0 1 0 0 1 0 0 0 0 0 1 0 1 0 1 0 1 0 1 0 1 1 0 0 0 0 0 1 0 0 1 0 0
0 1
0 0 1 0 1 0 1 0 0 0 0 0 1 0 1 0 0 1 0 1 0 0 1 0 0 0 1 0 1 0 1 0 1 0
1 0
1 0 0 0 0 0 0 0 0 1 0 1 0 1 1 0 0 1 0 1 0 1 0 0 1 0 1 0 1 0 0 0 0 0
0 1
0 0 0 1 0 1 1 0 0 1 0 1 0]

print("Cluster Centers:")

print(model.cluster_centers_)

Cluster Centers:
[[ 4.29793023 80.28488372]
 [ 2.09433    54.75      ]]

plt.scatter(
    x[y_means == 0, 0],
    x[y_means == 0, 1],
    s=50,
    label='Cluster 1'
)

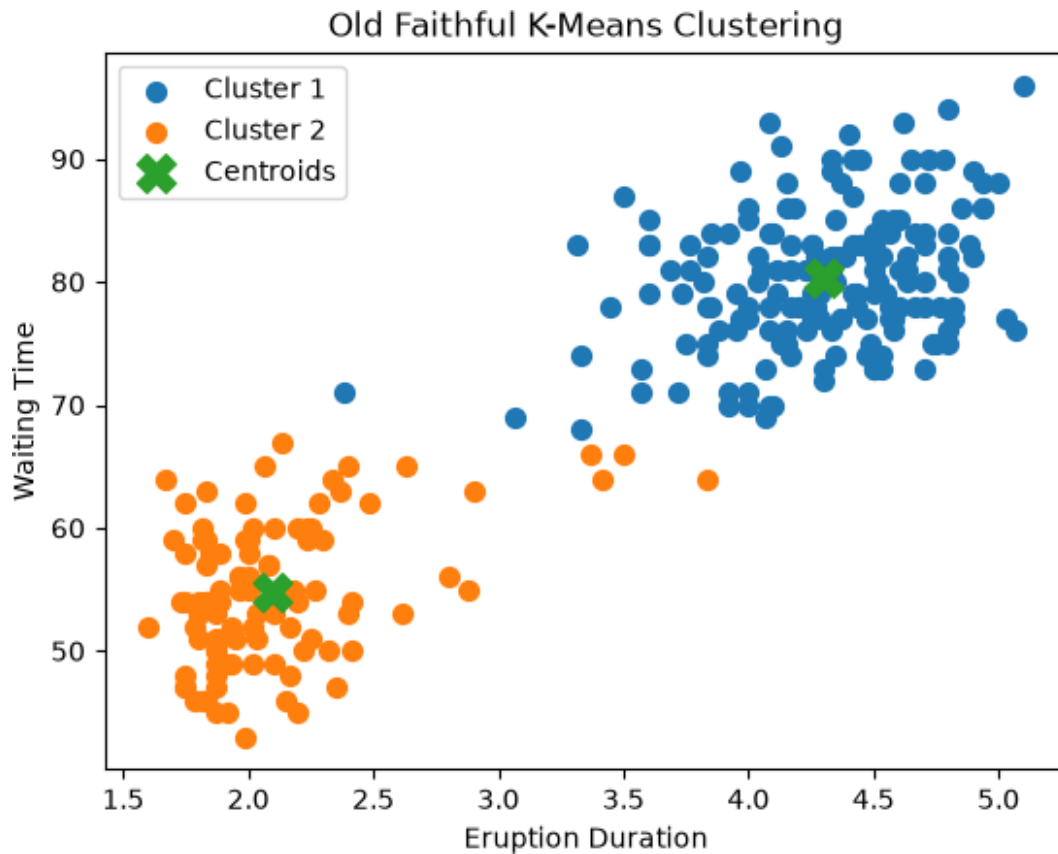
plt.scatter(
    x[y_means == 1, 0],
    x[y_means == 1, 1],
    s=50,
    label='Cluster 2'
)

plt.scatter(
    model.cluster_centers_[ :, 0],
    model.cluster_centers_[ :, 1],
    s=200,
    marker='X',
    label='Centroids'
)

plt.title('Old Faithful K-Means Clustering')
plt.xlabel('Eruption Duration')
plt.ylabel('Waiting Time')

```

```
plt.legend()
plt.show()
```



```
data['Cluster'] = y_means
```

```
print(data.head(10))
```

	eruptions	waiting	Cluster
0	3.600	79	0
1	1.800	54	1
2	3.333	74	0
3	2.283	62	1
4	4.533	85	0
5	2.883	55	1
6	4.700	88	0
7	3.600	85	0
8	1.950	51	1
9	4.350	85	0

```
data.to_csv(
    'old_faithful_kmeans_result.csv',
    index=False
)
```

```
print("Clustered dataset saved successfully!")
```

```
Clustered dataset saved successfully!
```